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PROJECT REPORT ON

NEXUS-SUPPORT

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2026-27

1. INTRODUCTION

Nexus-Support is a **decentralized offline disaster response and emergency communication system** designed for situations where conventional **cellular and Internet communication** may be unavailable. The system uses **Android devices as peer-to-peer communication nodes** to form a **local mesh network**. It enables users to exchange **emergency messages**, share **locations**, send **SOS alerts**, and access **emergency information** without requiring continuous Internet connectivity. The system uses technologies such as **Kotlin, Jetpack Compose, Google Nearby Connections, Room,** and **Android security services** to provide communication, storage, and secure data handling.

2. PROBLEM STATEMENT

During **natural disasters** and emergency situations, cellular networks and Internet services may become unavailable due to **infrastructure damage, network congestion, or power failures**. Existing communication systems that depend on **centralized servers** and continuous Internet connectivity may therefore become ineffective. A **decentralized system** is required to enable nearby devices to communicate, forward messages through intermediate devices, share emergency locations, and transmit **SOS information** even when conventional communication infrastructure is unavailable.

3. OBJECTIVES

- To develop an **Android-based decentralized offline emergency communication system**.
- To establish **peer-to-peer communication** between nearby Android devices.
- To implement **multi-hop mesh routing** for forwarding messages between devices.
- To provide **store-and-forward communication** when the destination device is temporarily unavailable.
- To provide **SOS alerts** and **emergency location sharing**.
- To provide **offline emergency guides** and essential information.
- To implement **secure communication** using encryption and cryptographic authentication.
- To optimize mesh communication using **battery-aware device scanning**.

4. METHODOLOGY & PROPOSED SYSTEM

The proposed system follows a **modular Android architecture** consisting of the **user interface, application logic, mesh communication, security,** and **local storage** components. The user interacts with the Android application through a **Jetpack Compose** interface. Nearby devices are discovered using **Google Nearby Connections** and participate as nodes in a **decentralized mesh network**. Messages are routed through available intermediate nodes using **multi-hop routing, hop limits, duplicate suppression,** and forwarding mechanisms. When a destination is unavailable, messages are stored locally and forwarded when connectivity becomes available. Emergency **SOS messages** and **location information** can be propagated through the mesh network. Secure communication is supported using **Android Keystore, cryptographic key exchange,** and **encrypted messaging**. Local data and pending messages are maintained using the device database so that essential operations can continue without Internet connectivity.

5. SYSTEM REQUIREMENTS

A. Software Requirements:

- **Operating System:** Windows 10/11 or Linux
- **IDE/Tools:** Android Studio
- **Programming Language:** Kotlin
- **Frontend:** Jetpack Compose
- **Communication:** Google Nearby Connections
- **Backend/Application Logic:** Kotlin, Coroutines, Hilt
- **Database:** Room / SQLCipher
- **Security:** Android Keystore / StrongBox
- **Mapping:** Mapsforge / Offline Maps

B. Hardware Requirements:

- **Processor:** Intel i3/i5 or equivalent AMD processor
- **RAM:** Minimum 4 GB / 8 GB
- **Storage:** 256/512 GB SSD / HDD
- **Android Smartphone** with Bluetooth, Wi-Fi, and GPS support

6. EXPECTED OUTCOME

The project will deliver a **functional Android-based offline disaster communication application** that enables users to communicate through a **decentralized mesh network** without continuous Internet connectivity. It will allow users to exchange messages, propagate **SOS alerts**, share locations, **store and forward messages**, and access emergency information securely and efficiently during disaster situations.